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Format ISO/DIN A3

Purpose of Document				Type of Document
Information		for Approval		As Built

Place for digital signature of Contractor

Place for digital signature of Revisor

AS BUILT

Rev.	Revision Description	Drawn by	Checked by	Approved by Designer	Approved by Contractor	Date
00	As Built	Antonio Aleksić	Grga Kolić	Zvonimir Lažeta		30.10.2026.

Document No. (CSD): PER-E-8-U6-SCH-001-00-VHHR

Employer:	Consultant:	Contractor:
epcg Elektroprivreda Crne Gore	ENERGOPROJEKT HIDROINŽENJERING a.d.	VOITH VOITH HYDRO GmbH&CoKG St. Pölten

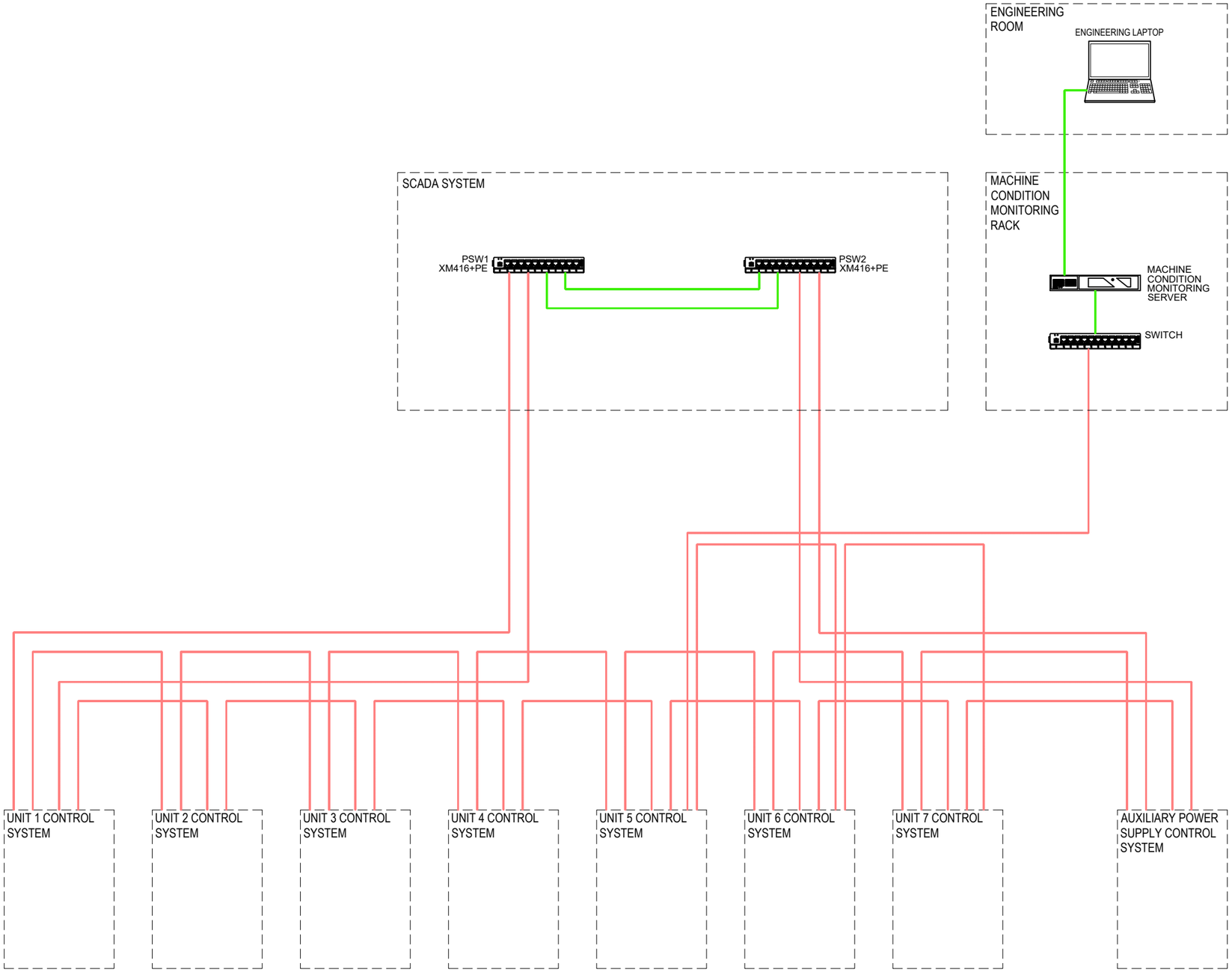
Project:

Reconstruction and Modernisation of HPP Perucica - Phase II

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica – Phase II	epcg Elektroprivreda Crne Gore AD	SINTAKSA A Voith Company, SINTAKSA d.o.o., Brza cesta 4, 21210 Mravince-Solin, Croatia
Approved by Designer	Zvonimir Lažeta	30.10.2026.			
Checked by	Grga Kolić	30.10.2026.			
Drawn by	Antonio Aleksić	30.10.2026.	PROJECT NAME	EMPLOYER	DESIGNER

CONTROL SYSTEM TOPOLOGY	DRAWING No.: PERU-01-16-170-12-001	REV No.: 00	=	
			+	
COVER SHEET	Designer No.:	SALES No.:	PAGE	1
			FOLLOW	2

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica – Phase II	epcg Elektroprivreda Crne Gore AD	SINTAKSA A Veolia Company	CONTROL SYSTEM TOPOLOGY	DRAWING No.: PERU-01-16-170-12-001	REV No.: 00	=
Approved by Designer	Zvonimir Lažeta	30.10.2026.			SINTAKSA d.o.o., Brza cesta 4, 21210 Mravice-Solin, Croatia				+
Checked by	Grga Kolić	30.10.2026.					Designer No.:	SALES No.:	PAGE 2
Drawn by	Antonio Aleksić	30.10.2026.	PROJECT NAME	EMPLOYER	DESIGNER	OVERVIEW			FOLLOW 5

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SCADA RING 1
SCADA RING 2
MACHINE CONDITION MONITORING NETWORK
PROTECTION RING
PROTECTION RING

IEC60870-5-104
IEC60870-5-104
IEC60870-5-104 / IEC 61850
IEC60870-5-104 / IEC 61850

+U5(6,7)CBC10
TURBINE GOVERNOR
CPU MASTER 1515-2 PN
CPU BACKUP 1515-2 PN
Profinet ring
DISTRIBUTED I/O MODULES ET200SP
CPU COMMON I/O MODULES 1510-1 PN
DISTRIBUTED I/O MODULES ET200SP
CWS
ET200AL
Multimeter SIEMENS PAC3220
HMI SIMATIC MTP1500 15.6"
Multimeter SIEMENS PAC3220

+U5(6,7)CJA10
UNIT CONTROL
DISTRIBUTED I/O MODULES ET200SP
MANUAL CONTROL
Multimeter SIEMENS PAC3220
Synchronizer WOODWARD
ALARM PANEL
SCALANCE XC216
SCALANCE XC216
IEC60870-5-104
IEC60870-5-104
IEC60870-5-104 / IEC 61850
CPU 1515-2 PN
IEC60870-5-104 / IEC 61850
HMI iEi 21,5" PPC2 IMP VIEW4 APPLICATION
DISTRIBUTED I/O MODULES ET200SP
Profinet ring

+U5(6,7)CMB10
MACHINE CONDITION MONITORING
Modbus TCP
NI CRIO
PARTIAL DISCHARGE ANALYSER
CALYSTO TRAFO MONITORING
SHAFT CURRENT MONITORING SYSTEM
Ethernet switch
HMI HIS-ML17-CTTH 17"

+U5(6,7)CHA10
EL. PROTECTION 1
MAIN generator protection relay SIEMENS SIPROTEC 7UM85
IEC 61850

+U5(6,7)CHB10
EL. PROTECTION 2
BACKUP generator protection relay SIEMENS SIPROTEC 7UM85
IEC 61850
IEC 61850

IEC60870-5-104
IEC60870-5-104

+U5(6,7)MKC10
EXCITATION
CPU 1517-3 PN/DP + ET200MP I/O MODULES
CPU 1517-3 PN/DP + ET200MP I/O MODULES
Profinet
Profinet
HMI SIMATIC MTP1500 15.6"

+U5(6,7)BJA10
400 VAC
Multimeter SIEMENS PAC3220
Multimeter SIEMENS PAC3220

GENERATOR
- Generator heaters
- Generator neutral point zone sensors
- Generator bearing cooling

DRAINAGE SYSTEM

STEP-UP TRANSFORMER

110kV GENERATOR BAY

LEGEND:
- Ethernet / Profinet / Modbus TCP-IP/ IEC60870-5-104
- Cu (Hard-wired)
- FO
- Profibus

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica – Phase II	epcg Elektroprivreda Crne Gore AD	SINTAKSA A Veolia Company SINTAKSA d.o.o., Brza cesta 4, 21210 Mravice-Solin, Croatia	CONTROL SYSTEM TOPOLOGY	DRAWING No.: PERU-01-16-170-12-001	REV No.: 00	=
Approved by Designer	Zvonimir Lažeta	30.10.2026.							+
Checked by	Grga Kolić	30.10.2026.					Designer No.:	SALES No.:	PAGE 3
Drawn by	Antonio Aleksić	30.10.2026.	PROJECT NAME	EMPLOYER	DESIGNER	UNITS A5, A6 & A7			FOLLOW 5

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SCADA RING 1
SCADA RING 2
PROTECTION RING
PROTECTION RING
IEC60870-5-104
IEC60870-5-104
IEC60870-5-104 / IEC 61850
IEC60870-5-104 / IEC 61850

+U1(2,3,4)CBC10
TURBINE GOVERNOR
CPU MASTER 1515-2 PN
CPU BACKUP 1515-2 PN
Profinet ring
DISTRIBUTED I/O MODULES ET200SP
CPU COMMON I/O MODULES 1510-1 PN
DISTRIBUTED I/O MODULES ET200SP
CWS
ET200AL
Multimeter SIEMENS PAC3220
HMI SIMATIC MTP1500 15.6"
Multimeter SIEMENS PAC3220

+U1(2,3,4)CJA21
UNIT CONTROL
SCALANCE XC216
SCALANCE XC216
IEC60870-5-104
IEC60870-5-104
IEC60870-5-104 / IEC 61850
IEC60870-5-104 / IEC 61850
CPU 1515-2 PN
HMI iEi 21,5" PPC2 IMP VIEW4 APPLICATION
Multimeter SIEMENS PAC3220
DISTRIBUTED I/O MODULES ET200SP
Profinet ring

+U1(2,3,4)CJA22
UNIT CONTROL
DISTRIBUTED I/O MODULES ET200SP

+U1(2,3,4)CHA10
EL. PROTECTION 1
MAIN generator protection relay SIEMENS SIPROTEC 7UM85
IEC 61850

+U1(2,3,4)CHB10
EL. PROTECTION 2
BACKUP generator protection relay SIEMENS SIPROTEC 7UM85
IEC 61850
IEC 61850

IEC60870-5-104
IEC60870-5-104

+U1(2,3,4)MKC10
EXCITATION
CPU 1517-3 PN/DP + ET200MP I/O MODULES
CPU 1517-3 PN/DP + ET200MP I/O MODULES
Profinet
Profinet
HMI SIMATIC MTP1500 15.6"

+U1(2,3,4)AEA10
GENERATOR BAY

LEGEND:
Ethernet / Profinet / Modbus TCP-IP/ IEC60870-5-104
Cu (Hard-wired)
FO
Profibus

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica – Phase II	epcg Elektroprivreda Crne Gore AD	SINTAKSA A Veolia Company SINTAKSA d.o.o., Brza cesta 4, 21210 Mravice-Solin, Croatia	CONTROL SYSTEM TOPOLOGY	DRAWING No.: PERU-01-16-170-12-001	REV No.: 00	=
Approved by Designer	Zvonimir Lažeta	30.10.2026.							+
Checked by	Grga Kolić	30.10.2026.					Designer No.:	SALES No.:	PAGE 4
Drawn by	Antonio Aleksić	30.10.2026.	PROJECT NAME	EMPLOYER	DESIGNER	UNITS A1, A2, A3 & A4			FOLLOW 5

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SCADA RING 1
SCADA RING 2
IEC60870-5-104 / IEC 61850
IEC60870-5-104 / IEC 61850

+UACJA10
AUX UNIT 1 CONTROL
IEC60870-5-104
IEC60870-5-104
SCALANCE XB008
SCALANCE XB008
CPU 1515-2 PN
IEC60870-5-104
IEC60870-5-104
Profinet ring
DISTRIBUTED I/O MODULES ET200SP
HMI SIMATIC MTP1500 15.6"
Multimeter SIEMENS PAC3220

+UV01
VIBRACONTROL
SCALANCE XC216
SCALANCE XC216
IEC60870-5-104
IEC60870-5-104 / IEC 61850
IEC60870-5-104 / IEC 61850
HMI iEi 21,5" PPC2 IMP VIEW4 APPLICATION

+UBCJA10
AUX UNIT 2 CONTROL
IEC60870-5-104
IEC60870-5-104
SCALANCE XB008
SCALANCE XB008
CPU 1515-2 PN
IEC60870-5-104
IEC60870-5-104
Profinet ring
DISTRIBUTED I/O MODULES ET200SP
HMI SIMATIC MTP1500 15.6"
Multimeter SIEMENS PAC3220

PROTECTION RING
IEC 61850
PROTECTION RING
IEC 61850

+UAMKC10
AUX UNIT 1 EXCITATION
IEC60870-5-104
IEC60870-5-104
CPU 1517-3 PN/DP + ET200MP I/O MODULES
Profinet
HMI SIMATIC MTP1500 15.6"

+?
PENSTOCK
AUMA K1:C1:V1

+?
PENSTOCK
AUMA K1:C2:V2

+UACHA10
AUX UNIT 1 EL. PROTECTION
Generator protection relay SIEMENS SIPROTEC 7UM85

+UBCHA10
AUX UNIT 2 EL. PROTECTION
Generator protection relay SIEMENS SIPROTEC 7UM85

+?
PENSTOCK
AUMA K2:C1:V1

+?
PENSTOCK
AUMA K2:C2:V2

+UBMKC10
AUX UNIT 2 EXCITATION
IEC60870-5-104
IEC60870-5-104
CPU 1517-3 PN/DP + ET200MP I/O MODULES
Profinet
HMI SIMATIC MTP1500 15.6"

LEGEND:
- Ethernet / Profinet / Modbus TCP-IP/ IEC60870-5-104
- Cu (Hard-wired)
- FO
- Profibus

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica – Phase II	epcg Elektroprivreda Crne Gore AD	SINTAKSA A Veltlin Company SINTAKSA d.o.o., Brza cesta 4, 21210 Mravice-Solin, Croatia	CONTROL SYSTEM TOPOLOGY UNITS KA1 & KA2	DRAWING No.: PERU-01-16-170-12-001	REV No.: 00	=
Approved by Designer	Zvonimir Lažeta	30.10.2026.							+
Checked by	Grga Kolić	30.10.2026.					Designer No.:	SALES No.:	PAGE 5
Drawn by	Antonio Aleksić	30.10.2026.	PROJECT NAME	EMPLOYER	DESIGNER				FOLLOW 5

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